

3. $x, y \in \mathbb{Z}$

$$\Rightarrow \lceil \frac{x}{y} \rceil = \begin{cases} \lceil \frac{x+y-1}{y} \rceil, & y > 0 \\ \lfloor \frac{x+y+1}{y} \rfloor, & y < 0 \end{cases}$$

$$= \begin{cases} \text{fix}(\frac{x+y-1}{y}), & x, y > 0 \\ \text{fix}(\frac{x+y+1}{y}), & x, y < 0 \end{cases}$$

$$\text{fix}(\frac{x}{y}), \quad x, y \leq 0$$

$$(\because \frac{x+y-1}{y} > 0, \text{ if } x, y > 0)$$

$$(\because \frac{x+y+1}{y} > 0, \text{ if } x, y < 0)$$

$\text{fix}(x) := \text{truncate } x \text{ towards } 0.$

$$= k+1 + \lceil \frac{r-1}{y} \rceil$$

$$= k+1$$

$$(\because r-1 \in [0, y-1])$$

$$\text{1.3 eg, } \lceil \frac{-7}{3} \rceil = -2 = \lceil \frac{-7+3-1}{3} \rceil$$

2. $x, y \in \mathbb{Z}, y < 0$

$$\Rightarrow \lceil \frac{x}{y} \rceil = \lfloor \frac{x+y+1}{y} \rfloor$$

$$\text{proof: let } x' = -x, y' = -y$$

$$\therefore \lceil \frac{x}{y} \rceil = \lceil \frac{x}{y'} \rceil = \lfloor \frac{x+y'+1}{y'} \rfloor$$

$$= \lfloor \frac{-x-y-1}{-y} \rfloor$$

$$= \lfloor \frac{x+y+1}{y} \rfloor$$

$$\text{2.3 eg, } \lceil \frac{7}{-3} \rceil = -2 = \lfloor \frac{7-3+1}{-3} \rfloor$$

$$\boxed{\lceil \frac{x}{y} \rceil, x, y \in \mathbb{Z}}$$

1. $x, y \in \mathbb{Z}, y > 0$

$$\Rightarrow \lceil \frac{x}{y} \rceil = \lfloor \frac{x+y-1}{y} \rfloor$$

$$\text{proof: let } x = ky + r, \quad k, r \in \mathbb{Z}, \quad r \in [0, y)$$

$$\text{① if } r=0 \Rightarrow \lceil \frac{x}{y} \rceil = k$$

$$\lfloor \frac{x+y-1}{y} \rfloor = \lfloor k + \frac{y-1}{y} \rfloor = k$$

$$\text{② if } r \neq 0 \Rightarrow \lceil \frac{x}{y} \rceil = k+1$$

$$1 \leq r < y$$

$$\lfloor \frac{x+y-1}{y} \rfloor = \lfloor \frac{ky+r+y-1}{y} \rfloor$$

$$= \lfloor \frac{(k+1)y+r-1}{y} \rfloor$$